CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Cache Organization

119 440 682 19

Lecture No.- 4



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Recap of Previous Lecture



Topic Cache Write

Topic Write Through & Write Back

Topic Write Allocate

Topics to be Covered



- Topic: Cache Mapping
- Topic: Direct Mapping
- Topic: Tag
- Topic: Tag Directory

when CPU accesses mm then only system buses are used.

[NAT]

#Q. Size of data sent to main memory from CPU:

1. For write hit, when a write through cache is used? → 1 data size
2. For write miss, when a write through cache is used? → 1 data size
3. For write hit, when a write back cache is used? → Nothing
4. For write miss, when a write back cache is used? → Nothing

[MCQ]

#Q. Size of data sent from main memory to cache:

1. For write hit, when a write through cache is used? ⇒ Nothing
2. For write miss, when a write through cache is used? ⇒ Nothing
3. For write hit, when a write back cache is used? ⇒ Nothing
4. For write miss, when a write back cache is used? ⇒ block

[NAT]

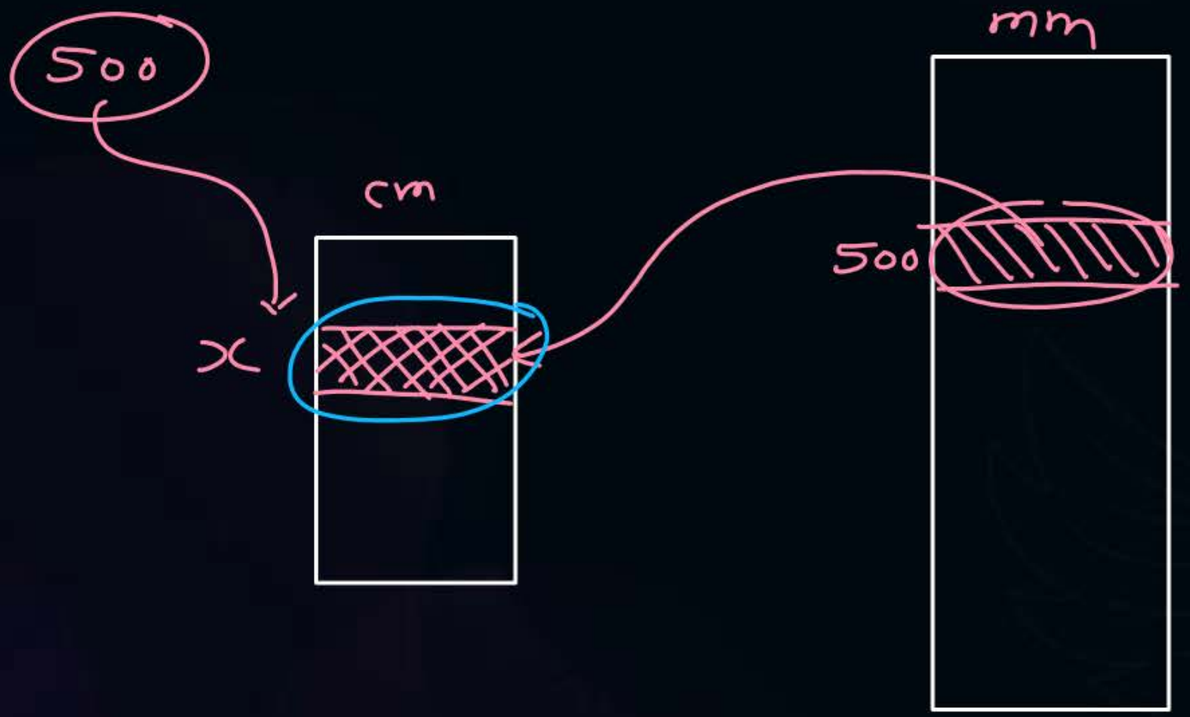
H. W.

#Q. Consider a system with average cycles per instruction (CPI) is 5 (without any operand access from memory). Consider the average memory access time is 5 cycles and 30% instructions only need 1 memory access for operand then the average CPI is _____ ?



Topic : Cache Mapping

CPU always generates mm address.



cm location

= fun (mm add.)

Cache mapping

There is a predefined pattern with the help of that content of mm from a specific address is to be brought to a specific location in cache.

This pattern is called as "cache mapping".

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mapping is applied on blocks.



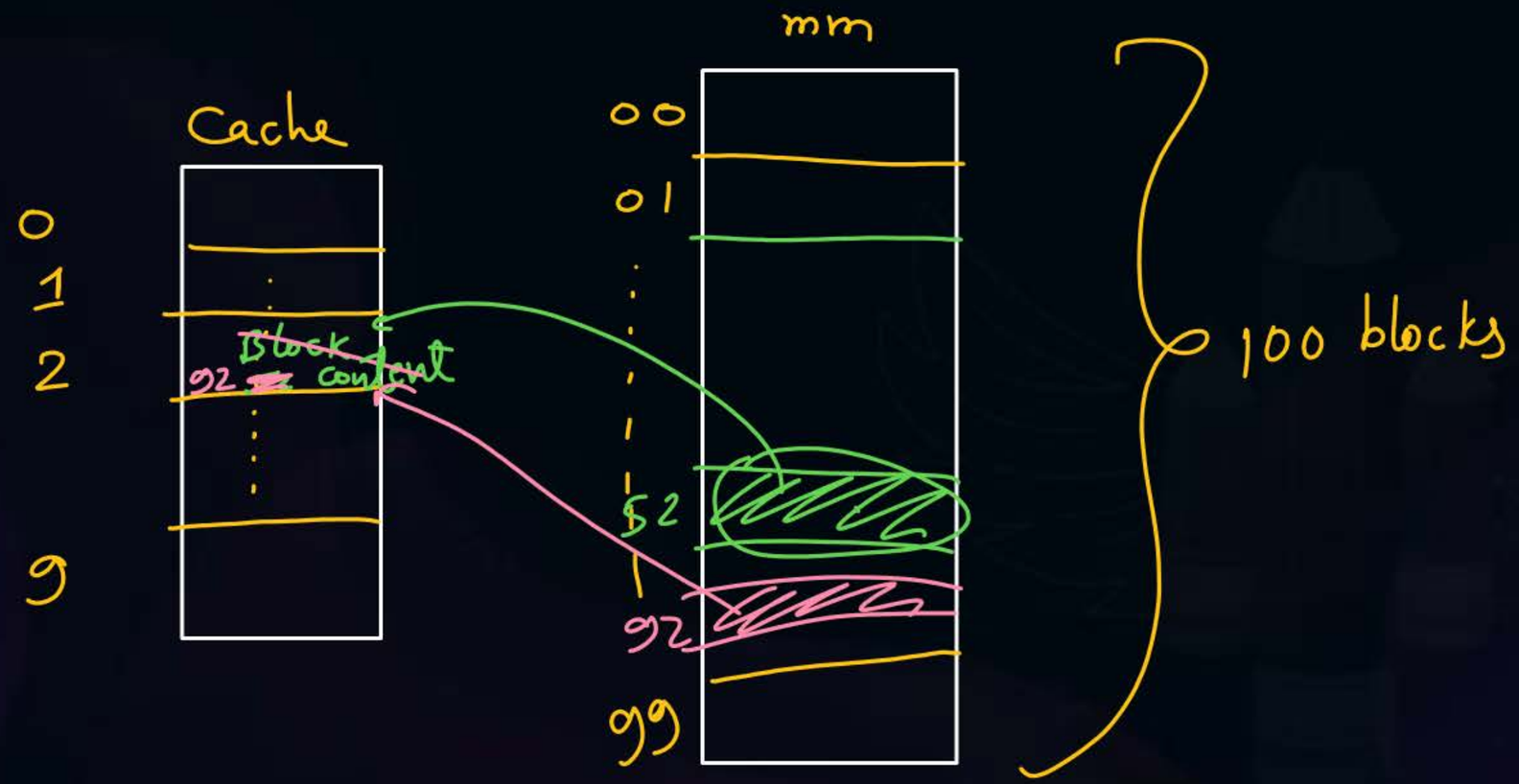
Topic : Cache Mapping

- Direct Mapping
- Set Associative Mapping
- Fully Associative Mapping



Topic : Direct Mapping

- Blocks in cache = 10 (0-9)
- Blocks in Main memory = 100 (00-99)





Topic : Direct Mapping

Cache	0	1	2	3	4	5	6	7	8	9
Main Memory	00	01	02	03	04	05	06	07	08	09
	10	11	12	13	14	15	16	17	18	19
	20	21	22	23	24	25	26	27	28	29
	30	31	32	33	34	35	36	37	38	39
	:	:	:	:	:	:	:	:	:	:
	90	91	92	93	94	95	96	97	98	99

35 % 10 = 5

38 % 10 = 8



Topic : Direct Mapping

cm block no. = (mm block no.) % (no. of blocks in cache)



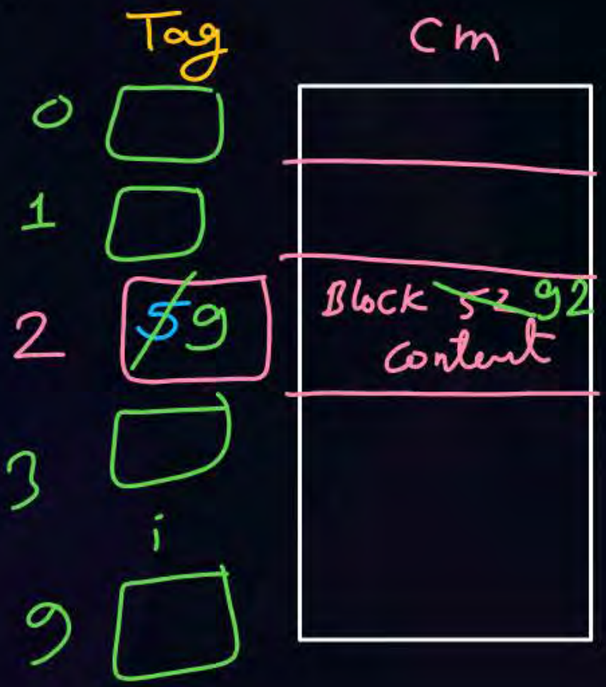
Topic : Direct Mapping

CPU Request (MM block)	Mapping (CM block no.)	Hit /Miss	Comments
52	= 52 % 10 = 2 CPU goes to cache at block 2. There is no any block in cache at block 2.	Miss	CPU bring mm block 52 content in cache, at block 2.
92	= 92 % 10 = 2 CPU goes to cm block 2 and checks there. Block 52 content is present there.	Miss	CPU brings mm block 92 content in cache, at block 2; by replacing block 52 content.



Topic : Direct Mapping

If any block is present in cache at some block, then to identify it that it is which main mem. block; Tag information is stored with it in cache.





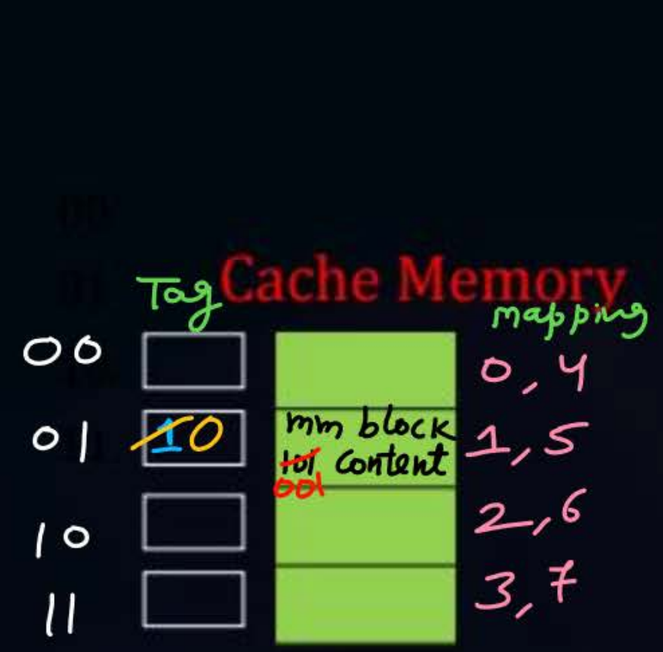
No. of tag informations stored in cache = no. of blocks in cache



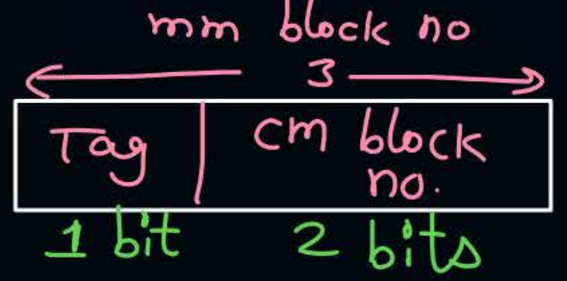
Topic : Direct Mapping

- Blocks in cache = 4 (00-11)
- Blocks in Main memory = 8 (000-111)

Topic : Direct Mapping

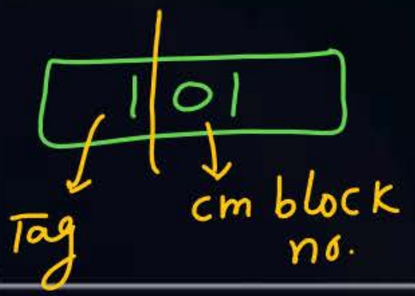
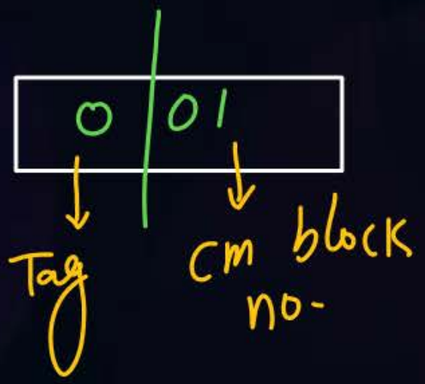








Topic : Direct Mapping

CPU Request (MM block)	Mapping (CM block no.)	Hit /Miss	Comments
$(5)_{10}$ $(101)_2$	5 % 4 = 1 	miss	bring mm block 101 content at cm block 01
$(1)_{10}$ $(001)_2$	1 % 4 = 1 	miss	bring mm block 001 content at cm block 01, by replacing block 101 content. update tag to 0.

no. of blocks in cache	no. of bits in cm block no.
4	2-bits
16	4-bits
2^{12}	12-bits
2^{x}	x bits
n	$\lceil \log_2 n \rceil$ bits



Topic : Direct Mapping

- Blocks in cache = 4 (00-11)
- Blocks in Main memory = 8 (000-111)
- Block Size = 2 Bytes
- Size of Cache memory = $4 * 2B = 8$ bytes
- Size of Main memory (byte addressable) = $8 * 2B = 16$ bytes $= 2^4$ Bytes
- Size of Main memory address = 4 bits

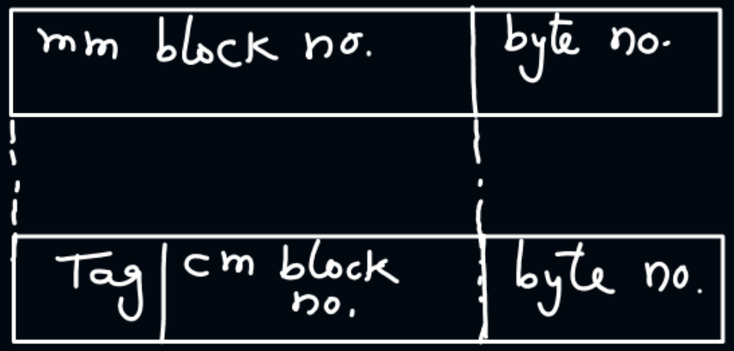


Topic : Direct Mapping



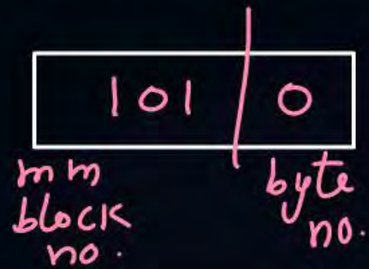
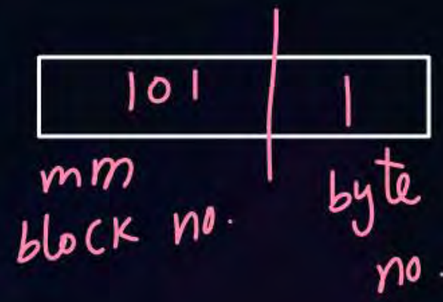


mm address



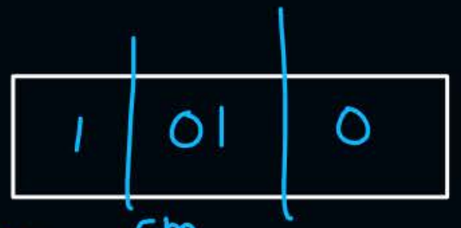


Topic : Direct Mapping

CPU Request (MM add.)	Mapping(CM block no.)	Hit/Miss	Comments
1010  	 	miss	bring mm block 101 content at cache block 01 with tag 1
1011  		Hit	CPU accesses byte 1 of this block from cache.



Topic : Direct Mapping



⇓

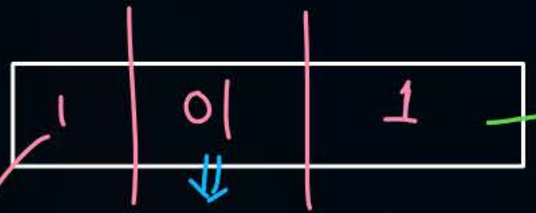
CPU goes to cm block 01

⇓

no any block there

⇓

miss



⇓

CPU goes to cm block 01.

⇓

check what is tag in cache ⇒ 1

Tag in CPU's Reg ⇒ 1

⇓

match

⇓

Hit

byte 1 of this block is sent to CPU.



2 mins Summary

Topic Cache Mapping

Topic Direct Mapping

Topic Tag

Topic Tag Directory



Happy Learning

THANK - YOU